

AI Smart Cafe – Final Project Proposal

Project Title

AI Smart Cafe – An Intelligent Cafe Website with AI-Powered Food & Beverage Recommendations

Project Overview

AI Smart Cafe is a modern, AI-powered web application designed to enhance the customer experience of a café. Unlike a traditional café website that only displays menus and contact information, AI Smart Cafe acts as a virtual café assistant. It helps customers discover the perfect food and drink based on their preferences, dietary needs, budget, mood, and occasion.

The application combines a professional café website with an AI assistant that provides personalized recommendations, making ordering easier and more enjoyable while helping customers make informed decisions.

This project demonstrates how Artificial Intelligence can improve customer service in the food and beverage industry while providing a complete end-to-end web application.

Problem Statement

Many café customers experience difficulties when choosing what to order because:

- They are unfamiliar with the menu.
- They have dietary restrictions.
- They have a limited budget.
- They don't know which food pairs best with certain drinks.
- They are ordering for groups or events.
- They are overwhelmed by too many menu choices.

Traditional café websites simply list menu items without providing guidance or personalization.

AI Smart Cafe solves this problem by acting as an intelligent digital waiter that recommends suitable menu items based on each customer's needs.

Proposed Solution

Develop a responsive AI-powered café website where visitors can:

- Browse the café menu.
- View food categories.

- Reserve tables.
 - Contact the café.
 - Explore the gallery.
 - Chat with an AI Cafe Assistant.
 - Receive personalized menu recommendations.
 - Plan meals within a budget.
 - Discover food and drink pairings.
 - Get dietary-friendly suggestions.
 - Receive event catering recommendations.
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Target Users

- Café customers
 - Students
 - Families
 - Tourists
 - Coffee lovers
 - Office workers
 - Event organizers
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Project Objectives

The main objectives of AI Smart Cafe are:

- Create a modern café website.
 - Improve customer ordering experience.
 - Demonstrate AI integration in a real-world business.
 - Provide personalized recommendations.
 - Build a complete deployed web application.
 - Showcase practical AI implementation.
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Core Features

1. Home Page

- Hero Banner
 - Featured Drinks
 - Today's Special
 - AI Assistant Button
 - Customer Reviews
 - Call to Action
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2. Menu Page

Categories:

- Coffee
- Tea
- Cold Drinks
- Smoothies
- Cakes
- Desserts
- Sandwiches
- Burgers
- Pizza
- Breakfast
- Snacks

Each menu item includes:

- Image
 - Name
 - Description
 - Price
 - Calories
 - Ingredients
 - Availability
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3. AI Cafe Assistant ★

This is the main feature of the application.

Customers can ask questions naturally.

Example:

I want something sweet under Rs.700.

AI:

I recommend the Caramel Latte with Chocolate Brownie because both are sweet and your total is Rs.650.

Example:

I'm vegetarian.

AI suggests only vegetarian menu items.

Example:

I don't like coffee.

AI recommends tea, smoothies, juices, or desserts.

4. Smart Budget Planner

Customer enters:

- Budget
- Number of people

AI generates:

- Complete meal
- Drinks
- Total cost

Example:

Budget:

Rs.2500

People:

2

Recommendation:

- 2 Cappuccinos
- Chicken Alfredo Pasta
- Chocolate Lava Cake

Estimated Cost:

Rs.2450

5. Food Pairing Recommendation

Example:

Customer chooses:

Americano

AI suggests:

- Chocolate Cookie
- Cheesecake
- Brownie

With reasons.

6. Dietary Recommendation

Supports:

- Vegetarian
- Vegan
- Halal
- Gluten-Free
- Low Sugar
- High Protein

AI filters suitable menu items.

7. Event Planner

Customer enters:

- Birthday
- Meeting
- Family Dinner
- Friends Gathering

AI recommends:

- Food quantity
 - Drinks
 - Dessert
 - Estimated budget
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8. Reservation System

Customers can:

- Select date
- Select time
- Number of guests
- Phone Number
- Email

Reservation confirmation is stored in the database.

9. Contact Page

Includes:

- Google Map
 - Contact Form
 - Phone Number
 - Email
 - Social Media Links
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10. Gallery

Display:

- Food Images
 - Coffee Images
 - Interior
 - Outdoor Seating
 - Customer Moments
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11. Reviews

Visitors can:

- Give ratings
 - Write reviews
 - View customer testimonials
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12. Admin Dashboard (Optional)

Admin can:

- Add Menu Items
 - Delete Menu Items
 - Update Prices
 - View Reservations
 - View Customer Reviews
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AI Features

The AI assistant will understand customer preferences and provide personalized suggestions.

It will consider:

- Budget
 - Food Preferences
 - Favorite Drinks
 - Sweet/Bitter Taste
 - Vegetarian Options
 - Allergies
 - Weather
 - Time of Day
 - Group Size
 - Meal Type
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AI System Prompt

The AI assistant will operate using a carefully designed system prompt similar to the following:

You are the AI assistant for AI Smart Cafe. Your role is to help customers choose food and beverages from the café menu. Recommend only menu items available in the database. Consider customer budget, dietary preferences, taste, meal type, weather, and group size. Explain every recommendation in a friendly and concise manner. Never invent menu items or prices. If information is unavailable, politely state that instead of making assumptions.

Website Pages

- Home
 - Menu
 - AI Assistant
 - Reservation
 - Gallery
 - Reviews
 - About
 - Contact
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Technologies Used

Frontend

- Next.js

- React
- Tailwind CSS

Backend

- Supabase (or Firebase)

Database

- PostgreSQL (Supabase)

AI

- Google Gemini API

Deployment

- Vercel

Version Control

- GitHub
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Database Tables

Menu

- id
- name
- category
- description
- ingredients
- price
- calories
- dietary_tags
- image_url
- available

Reservations

- id
- customer_name
- email
- phone
- guests
- reservation_date
- reservation_time
- notes

Reviews

- id
 - customer_name
 - rating
 - comment
 - created_at
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User Flow

1. User opens the website.
 2. Browses the menu.
 3. Opens the AI Assistant.
 4. Describes preferences.
 5. AI recommends suitable food and drinks.
 6. User reserves a table.
 7. User visits the café.
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Expected Outcomes

The application will:

- Improve customer experience.
 - Simplify menu selection.
 - Demonstrate practical AI integration.
 - Provide personalized recommendations.
 - Showcase a complete AI-powered business website.
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Future Enhancements

- Online payment integration
 - Real-time order tracking
 - Customer loyalty program
 - QR code ordering
 - Voice-enabled AI assistant
 - Multilingual support (English and Urdu)
 - AI-generated daily specials
 - Inventory-aware recommendations
 - Personalized user accounts and order history
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Conclusion

AI Smart Cafe is more than a digital menu—it is an intelligent café companion that combines a professional web experience with Artificial Intelligence. By helping customers choose meals based on their tastes, dietary needs, budgets, and occasions, the application demonstrates a practical, user-centered use of AI. The project is designed to be fully functional, visually appealing, and deployable, making it a strong example of an end-to-end AI application suitable for the ACT AI Final Project.